

ATVED GridLock - Official Demo Guide

Flickar GridLock Road Safety Hackathon

1. How to Host the Backend & Frontend Locally

You need to open TWO separate terminals to host the project locally on your laptop.

TERMINAL 1 (The Backend API & Database):

```
cd Desktop/GridLock
pip install -r requirements.txt
python setup_db.py
python api_server.py
```

This runs the FastAPI server on port 8000 and creates the ATVED database.

TERMINAL 2 (The Web Application UI):

```
cd Desktop/GridLock
python -m http.server 8080 -d dashboard
```

This serves your beautiful HTML/CSS frontend. Open your web browser and go to <http://localhost:8080>

2. How to Perform the Demo for the Judges

Follow this exact sequence to blow the judges away:

Step 1: Show the Citizen Dashboard (The 'Before' State)

- Have your teammate log into the Citizen Portal (<http://localhost:8080>) using your Aadhaar: 4832 7651 9023
- Show the judges your perfect Traffic Score (1000), 1.0x multiplier, and Bank Balance (Rs. 50,000).

Step 2: Trigger the Live Computer Vision AI

- Tell the judges: 'We are now feeding live traffic camera footage to our YOLO model.'
- Open a THIRD terminal and run the model on your downloaded video file:

```
python computer_vision_models/demo_live_end_to_end.py your_video.mp4
```

- The model window will pop up drawing red bounding boxes on violators. Behind the scenes, the

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script automatically overrides the random license plate with YOUR license plate (AS 01 KK 4521) to prove the end-to-end flow.

Step 3: The Magic Reveal (The 'After' State)

- Switch back to the Citizen Portal in your browser and hit Refresh.
- The judges will instantly see your Traffic Score dropped, your bank account was automatically deducted, and a new fine appeared in your history table!
- Switch to the Authority Control Room to show the Live Map and Fines counter ticking up.

Step 4: The Official E-Challan Receipt

- Finally, open the 'evidence/challans/' folder in your project.
- Open the newly generated PDF receipt to show the judges the official fine breakdown.